The Standard Normal Distribution and the Z-table

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Cumulative distribution function (CDF); Normal distribution.
- **Leads to:** Confidence intervals; Z-test and rejection regions.

1. Overview

Key point: Standardizing turns any normal distribution into the standard normal distribution $N(0, 1)$; one table of its areas, the **z-table** (G-2142), then answers probability questions about every normal variable.

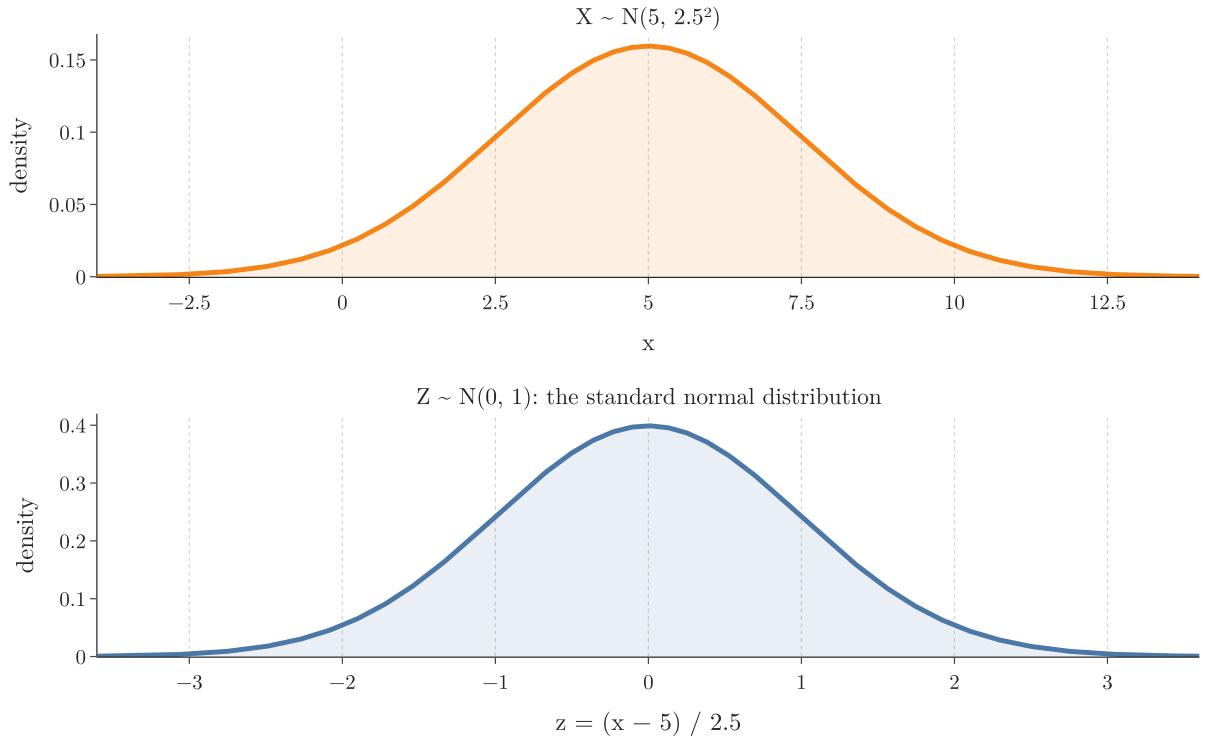


Figure 1: Standardizing: X ~ N(5, 2.5^2) becomes Z ~ N(0, 1); every value moves to its z-score

The notation $X \sim N(5, 2.5^2)$ reads "the variable X follows a normal distribution with mean 5 and variance 2.5^2", so its standard deviation is 2.5 (see parameters and notation).

Figure 1 shows the idea. The orange curve is a normal distribution with mean 5 and standard deviation 2.5. Subtracting 5 and dividing by 2.5 turns it into the blue curve, which has mean 0 and standard deviation 1. The shape stays exactly the same; only the numbers on the axis change.

This Note builds on what the normal distribution is. It covers:

- the standard normal distribution and how to standardize (sections 2 and 3);
- how to read a z-table (section 4);
- two problems solved with it (section 5);
- the **68-95-99.7 rule** (G-53), derived (section 6);
- where the normal distribution is used in data science (section 7).

2. The standard normal distribution

> **Key point:** The standard normal distribution is the normal distribution with mean 0 and standard deviation 1, written $Z \sim N(0, 1)$.

Among all normal distributions, one is special: the one with $\mu = 0$ and $\sigma = 1$. This distribution is called the **standard normal distribution** (G-1873; or **standard normal variate**) and is written with the letter Z:

$$Z \sim N(0, 1)$$

Its curve (Figure 1, bottom) is centred at 0, and its x axis counts standard deviations directly: 1 means one standard deviation above the mean, -2 two below.

1. **In words:** put $\mu = 0$ and $\sigma = 1$ into the normal PDF (probability density function: the formula for the height of the curve, see the PDF of the normal distribution).
2. **Formula:**
$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$
The standard normal PDF has its own symbol, ϕ (phi, G-19), and its CDF (cumulative distribution function: the area under the curve to the left of a point, see the CDF of the normal distribution) is written Φ (capital phi, G-18).
3. **Example:** at $z = 0$ the curve has its peak (Figure 2):
$$\phi(0) = 1/\sqrt{2\pi}$$
$$\phi(0) = 1/2.5066 = 0.3989$$
At $z = 1$:
$$\phi(1) = 0.3989 \times e^{-0.5}$$
$$\phi(1) = 0.3989 \times 0.6065 = 0.2420$$

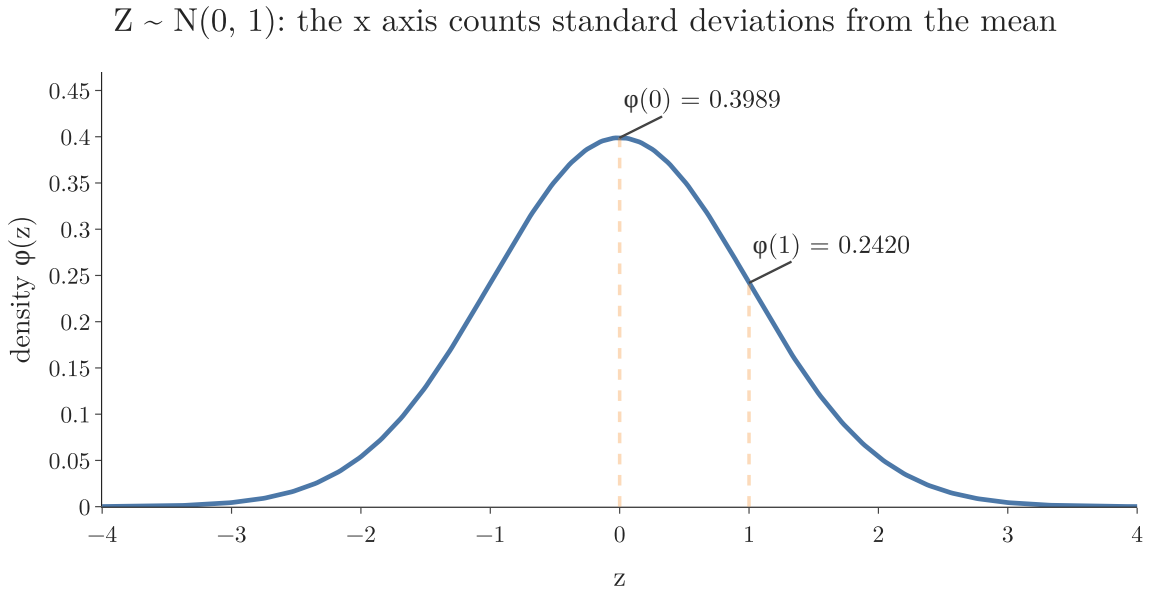


Figure 2: The standard normal PDF with the two worked values: the peak $\phi(0) = 0.3989$ and $\phi(1) = 0.2420$, one standard deviation to the right.

3. Standardizing a normal variable

> **Key point:** Subtracting the mean and dividing by the standard deviation turns $X \sim N(\mu, \sigma^2)$ into $Z \sim N(0, 1)$.

To turn any normal variable into a standard normal one, we standardize every value: subtract the mean, divide by the standard deviation. The result is its **z-score** (G-2141). Turning values into z-scores is the **standardization** (G-1874) of the standardization formula:

In plain words, a z-score is the number of standard deviations a value lies above or below the mean. A positive z-score is above the mean, a negative one below, and the further from 0, the more unusual the value. As a formula, with x the value, μ the mean and σ the standard deviation:

$$z = \frac{x - \mu}{\sigma}$$

For Figure 1, take $x = 10$ with mean 5 and standard deviation 2.5:

$$z = (10 - 5)/2.5 = 2$$

The value 10 lies two standard deviations above the mean. Every tick on the top axis lines up with its z-score below.

Figure 3 shows the two steps as motion. Subtracting 5 slides the curve left until its centre sits at 0. Dividing by 2.5 squeezes it 2.5 times narrower, and the curve grows 2.5 times taller

so the total area stays 1. Watch the shaded tail: the cut-off rides from 10 to 2, and the area beyond it stays 0.0228 the whole time. That unchanged area is why $P(X > 10) = P(Z > 2)$, and why one table of Z serves every normal variable.

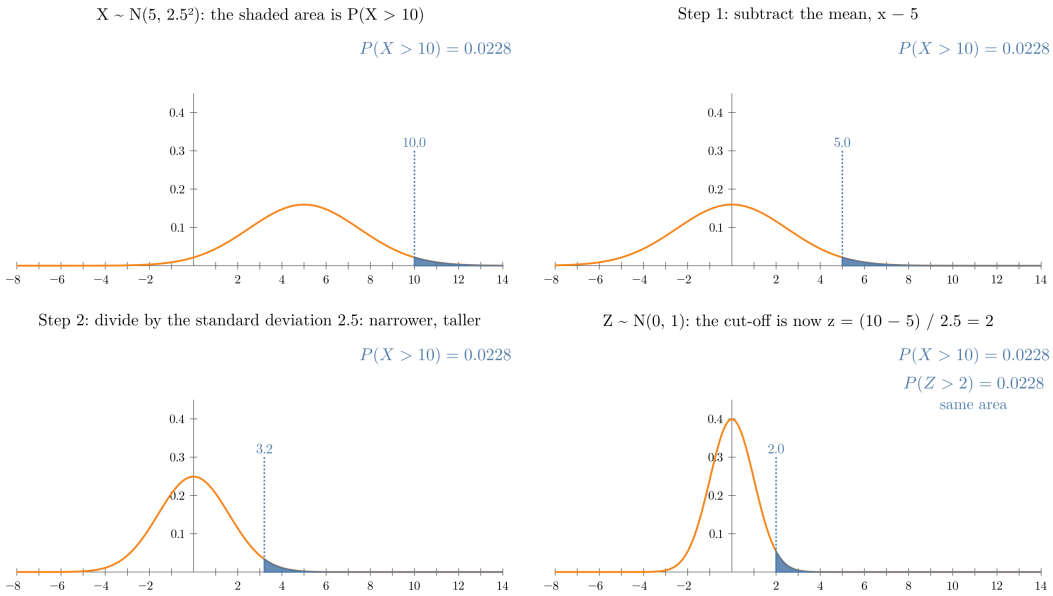


Figure 3: Standardizing as motion: X ~ N(5, 2.5^2) slides left by its mean, then squeezes by its standard deviation into N(0, 1); the shaded tail beyond 10, later beyond z = 2, keeps the same area 0.0228

Real data works the same way. A **feature** (G-772) is one variable of the data, one column of the table; an **observation** (G-1374) is one record, one row. The `Age` feature of the Titanic data (714 known ages) is roughly bell-shaped (a little skewed to the right, with a small bump of children; Figure 9), with mean 29.70 years and standard deviation 14.53 years. Standardizing it gives a feature with mean 2×10^{-16} (0 up to rounding) and standard deviation 1.

Python: Standardizing a feature by hand.

```
import pandas as pd

df = pd.read_csv("data/titanic_train.csv")
age = df["Age"].dropna()
z = (age - age.mean()) / age.std()
z.mean(), z.std()       # about 0, and 1
```

Standardizing changes the scale, not the shape: a skewed feature stays skewed (see each column keeps its shape). Only a normal feature becomes standard normal.

3.1 Why standardize?

> **Key point:** Standardizing lets us compare different normal distributions on one scale, and compute every probability from one table.

Two benefits make the standard normal distribution so useful:

1. **Comparison.** Two normal distributions with different means and spreads (heights in centimetres and in inches, marks in two different exams) can be compared side by side once both are standardized. For example, a student scores 80 in an exam with mean 70 and standard deviation 10, and 65 in another exam with mean 50 and standard deviation 5. The raw marks favour the first exam. The **z-scores** (G-2141) say the opposite:
$$z_1 = \frac{80 - 70}{10} = 1$$
$$z_2 = \frac{65 - 50}{5} = 3$$
The second result is three standard deviations above its mean, a far rarer performance than the first.
2. **One table for all probabilities.** The areas under the standard normal curve have been computed once and printed in a table. Any probability about any normal variable becomes a lookup in that table after standardizing.

4. The z-table

> **Key point:** A z-table lists $\Phi(z)$, the area under the standard normal curve to the left of z: the probability that Z is at most z.

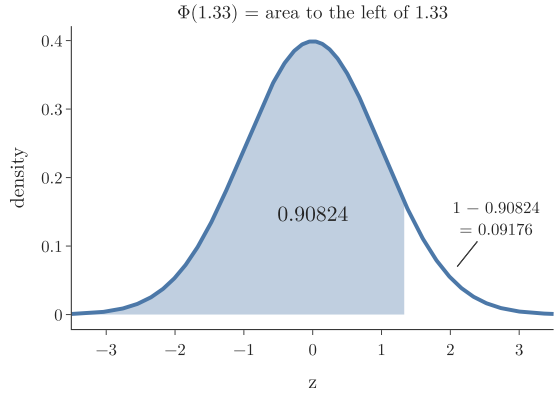
A **z-table** (G-2142), or standard normal table, lists the CDF of the standard normal distribution for many values of z:

$$\Phi(z) = P(Z \leq z) = \text{area under the curve from } -\infty \text{ to } z$$

The table gives an **area**, so a probability, not the height of the curve at z (Figure 4, left). There are two tables: one for negative z and one for positive z.

To look up a **z-score** (G-2141) with two decimals, such as 1.33:

1. Choose the positive table, because 1.33 is positive.
2. Find the row for the first two digits, **1.3**.
3. Find the column for the second decimal, **0.03**.
4. Read the cell where they cross: **0.90824** (Figure 4, right).



Positive z-table (excerpt)

z	0.00	0.01	0.02	0.03	0.04	0.05
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943

Figure 4: The area to the left of z = 1.33 (left) and where it sits in the z-table (right)

So $P(Z \leq 1.33) = 0.90824$: 90.8% of the area lies to the left of 1.33. The area to the right is the rest:

$$1 - 0.90824 = 0.09176$$

Python: The z-table in one line.

```
from scipy import stats

stats.norm.cdf(1.33)    # 0.90824, Phi(1.33)
stats.norm.sf(1.33)     # 0.09176, 1 - Phi(1.33)
stats.norm.ppf(0.975)   # 1.96, the z with 97.5% to its left
```

5. Two problems solved with the z-table

Key point: Standardize the value, look up its area, and subtract when the question asks for "more than" or "between".

5.1 Taller than 72 inches

Key point: About 9.2% of men are taller than 72 inches when heights are $N(68, 3^2)$.

The heights of adult men in a population are normal with mean 68 inches and standard deviation 3 inches. What is the probability that a randomly chosen man is taller than 72 inches?

1. **In words:** turn 72 into a **z-score** (G-2141), read the area to its left, and subtract it from 1 to get the area to its right.
2. **Formula:**

$$P(X > 72) = 1 - \Phi\left(\frac{72 - 68}{3}\right)$$

3. **Example:** the z-score is

$$z = \frac{72 - 68}{3} = \frac{4}{3} = 1.33$$

The **z-table** (G-2142) gives $\Phi(1.33) = 0.90824$, so

$$P(X > 72) = 1 - 0.90824 = 0.09176$$

About 9.2% of men are taller than 72 inches.

Figure 5 draws the problem. The shaded tail is the answer; the second row of tick labels shows that 72 inches sits at $z = 1.33$ on the standard normal scale.

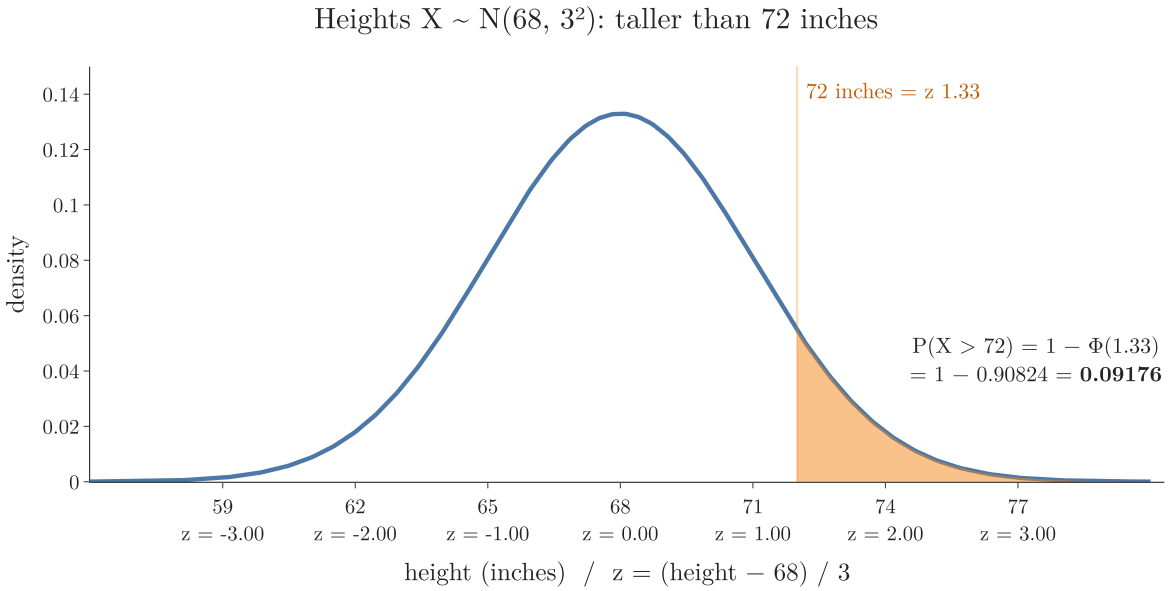


Figure 5: Heights N(68, 3^2). The area above 72 inches (orange) is $1 - \Phi(1.33) = 0.09176$; the tick labels give each height's z-score.

The steps are always the same: draw the curve, standardize the value, look up the area, then subtract if needed.

5.2 Between the mean and one standard deviation above it

> **Key point:** For every normal distribution, 34.13% of the values lie between μ and $\mu + \sigma$.

Now no numbers are given: $X \sim N(\mu, \sigma^2)$. What share of the values lies between the mean μ and $\mu + \sigma$?

1. **In words:** standardize both ends, then subtract the area left of the lower end from the area left of the upper end.
2. **Formula:**

$$z_{\text{lower}} = \frac{\mu - \mu}{\sigma} = 0$$

$$z_{\text{upper}} = \frac{(\mu + \sigma) - \mu}{\sigma} = 1$$

$$P(\mu \leq X \leq \mu + \sigma) = \Phi(1) - \Phi(0)$$

3. **Example:** the **z-table** (G-2142) gives $\Phi(1) = 0.8413$, and $\Phi(0) = 0.5$ because the curve is symmetric. So

$$P(\mu \leq X \leq \mu + \sigma) = 0.8413 - 0.5 = 0.3413$$

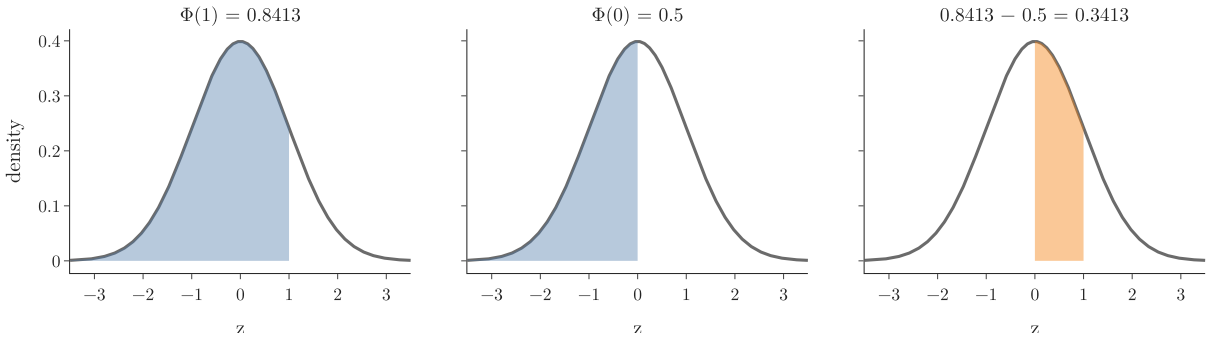


Figure 6: The area between 0 and 1 as the difference of two z-table areas

The answer, 34.13%, does not depend on μ or σ: it holds for every normal distribution.

6. Deriving the 68-95-99.7 rule

> **Key point:** Repeating the calculation for 1, 2 and 3 standard deviations, and doubling by symmetry, gives 68.27%, 95.45% and 99.73%.

Figure 7 repeats the calculation of section 5.2 for 1, 2 and 3 standard deviations. By symmetry, the same 34.13% lies between $\mu - \sigma$ and μ. So the share within one standard deviation on either side is

$$P(\mu - \sigma \leq X \leq \mu + \sigma) = 2 \times 0.3413 = 0.6827$$

The same steps for 2 and 3 standard deviations:

Range	Area right of the mean	Both sides
$\mu \pm 1\sigma$	$\Phi(1) - 0.5 = 0.8413 - 0.5 = 0.3413$	68.27%
$\mu \pm 2\sigma$	$\Phi(2) - 0.5 = 0.9772 - 0.5 = 0.4772$	95.45%
$\mu \pm 3\sigma$	$\Phi(3) - 0.5 = 0.99865 - 0.5 = 0.49865$	99.73%

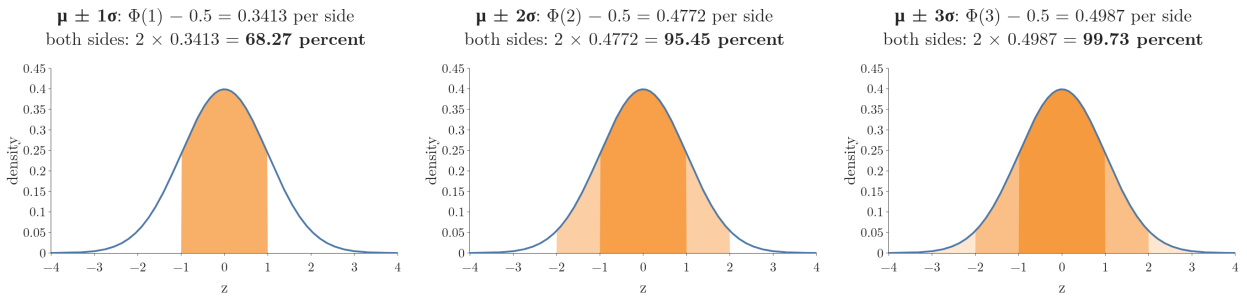


Figure 7: The areas within 1, 2 and 3 standard deviations of the mean, shaded one step at a time: 68.27, 95.45 and 99.73 percent.

The 68-95-99.7 pattern is the **empirical rule** (G-53) of the 68-95-99.7 rule, now derived from the **z-table** (G-2142) instead of taken on trust. The rule is powerful: knowing only that a variable is normal, without seeing any data, we can say that 99.73% of its values lie within 3 standard deviations of the mean.

A value far outside that range is extraordinary. Don Bradman's Test batting average of 99.94 is a famous example: it lies about 4.4 standard deviations above the mean of Test cricketers (Davis 2000). If the averages were normal, fewer than 1 value in 100,000 would lie that far up:

$$1 - \Phi(4.4) \approx 0.000005$$

Roughly speaking, most good batsmen sit within one or two standard deviations of the mean, and only the very greatest approach three.

6.1 Using the rule without a table

> **Key point:** The rule gives the middle; symmetry splits what is left over into two equal tails. Together they answer "below" and "above" questions in the head.

The rule can also be run in the other direction, to get areas quickly with no table. Figure 8 shows the steps with the rounded figures 68 and 95:

1. **The middle.** 68 percent of the values lie within one standard deviation of the mean.
2. **The leftover.** The rest, in percent:

$$100 - 68 = 32$$

   The curve is symmetric, so each tail holds half of it: 16 percent.

3. **Below $z = 1$.** Everything left of 1 is the middle plus the left tail, in percent:

$$68 + 16 = 84$$

The **z-table** (G-2142) agrees: $\Phi(1) = 0.8413$. Below $z = -1$ lies only the left tail, 16 percent.

4. **Above $z = 2$.** Within two standard deviations lie 95 percent, so 5 percent is left over, 2.5 percent in each tail. The z-table gives:

$$1 - \Phi(2) = 0.0228$$

The rule's round 95 makes the mental answer slightly high.

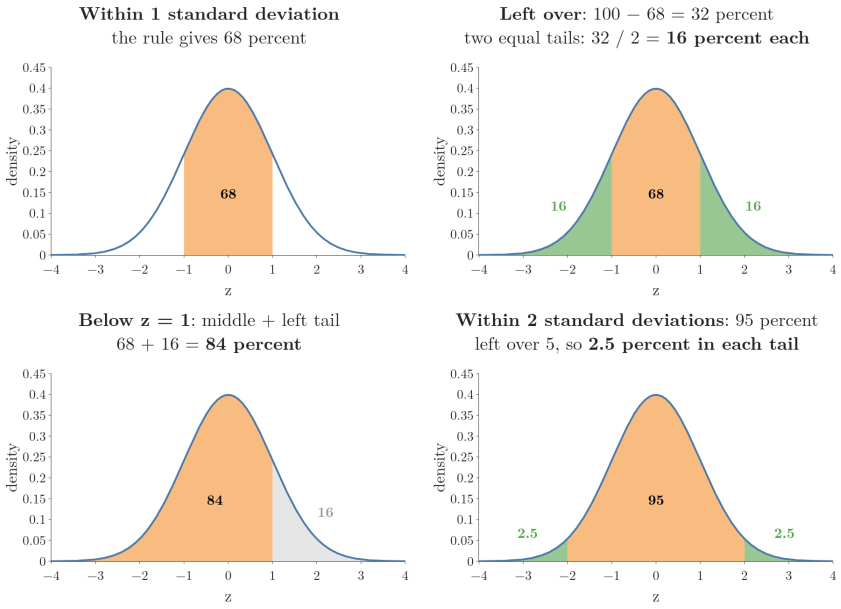


Figure 8: Tail areas from the rule: the middle 68 percent leaves 32, split into two tails of 16; below z = 1 lies 68 + 16 = 84 percent; beyond 2 standard deviations each tail holds 2.5 percent. Idea after Khan Academy, "ck12.org exercise: Standard normal distribution and the empirical rule"

7. Where the normal distribution is used in data science

> **Key point:** Outlier detection, the assumptions of some ML models, hypothesis testing, and the central limit theorem.

1. **Outlier detection.** For a **feature** (G-772) that is roughly normal, values beyond $\mu \pm 3\sigma$ are treated as outliers (values far from the rest; the z-score method, see the limits). For the Titanic ages, three standard deviations are

$$3 \times 14.53 = 43.59$$

so the limits are

$$29.70 - 43.59 = -13.89$$

$$29.70 + 43.59 = 73.29$$

(73.28 with the unrounded mean and standard deviation). No age is negative, so only the upper limit matters: two passengers, aged 74 and 80, are flagged as **outliers** (G-1420; Figure 9).

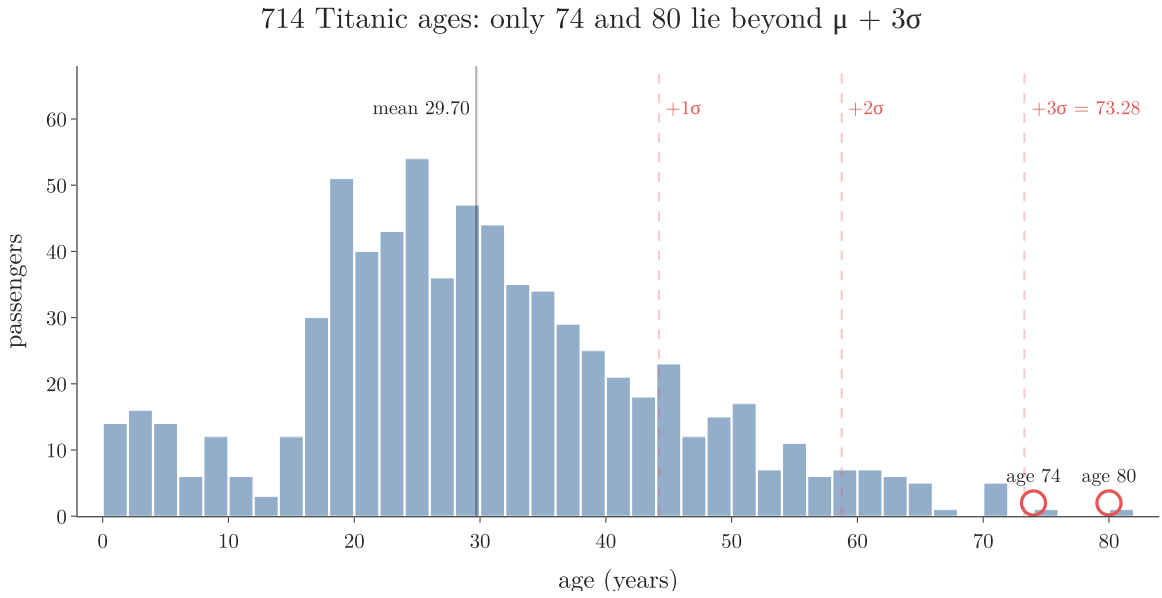


Figure 9: The 714 Titanic ages with the mean and the +1, +2 and +3 standard deviation lines. Only the passengers aged 74 and 80 (circled) lie beyond 73.28.

2. **Assumptions of ML models.** Some models perform better, or rely on the assumption, that something is normally distributed. Linear regression assumes that the **residuals** (G-705; the errors) are normal, not the inputs (see assumption 3: normal residuals). Linear and logistic regression also tend to work better on normal-looking inputs (see why make data normal), and a Gaussian mixture model (a density built by adding several normal curves) is built from normal curves (see **parametric density estimation** (G-1451)).

3. **Hypothesis testing** (G-913; checking a claim about a population with a sample). Many statistical tests assume that the data, or a statistic computed from it, is normally distributed.

4. **The central limit theorem** (G-364). Averages of samples from almost any distribution, normal or not, follow approximately a normal distribution, more closely as the samples grow (Pishro-Nik §7.1.2). This result, taught in the central limit theorem, is what makes the normal distribution central to inferential statistics.

8. Summary

Idea	Formula	Example	Why it matters
Standard normal	$Z \sim N(0,1)$	peak $\phi(0) = 0.3989$	one curve whose areas are printed once, in the z-table
Standardize	$z = (x - \mu)/\sigma$	$(72 - 68)/3 = 1.33$	keeps every area, so one table serves every normal variable
Z-table	$\Phi(z) = P(Z \leq z)$	$\Phi(1.33) = 0.90824$	turns a z-score into a probability by lookup
More than x	$1 - \Phi(z)$	$P(X > 72) = 0.092$	the table gives the left area, so subtract it from the total 1
Between two values	$\Phi(z_2) - \Phi(z_1)$	$\Phi(1) - \Phi(0) = 0.3413$	the left area up to z_2 minus the left area up to z_1 leaves the strip between
Empirical rule	$2(\Phi(k) - 0.5)$	68.27%, 95.45%, 99.73%	lets us answer "below" and "above" questions in the head, without a table

- A **z-table** (G-2142) gives areas (probabilities) to the left of z, not heights of the curve, so "more than" and "between" questions need a subtraction.
- Standardizing keeps the shape; only normal data becomes standard normal, so a skewed **feature** (G-772) stays skewed and the z-table does not give its probabilities.
- Values beyond 3 standard deviations are rare: 0.27% in all, so the z-score method flags them as outliers (the Titanic ages 74 and 80).
- This answers the opening question: standardize, then one table of $N(0,1)$ areas gives probabilities for every normal variable.

9. Sources

Built from

- CampusX, "Session 41 - Normal Distribution | DSMP 2023", YouTube, https://www.youtube.com/watc
- Khan Academy, "Z-score introduction", YouTube, https://www.youtube.com/watch?v=5S-Zfa-vOXs
- Khan Academy, "ck12.org exercise: Standard normal distribution and the empirical rule", YouTube, https://www.youtube.com/watch?v=2fzYE-Emar0

Other references

- Davis, C. (2000). *The Best of the Best: A Study of Batting and Bowling in Test Cricket*. ABC Books. Bradman's average 4.4 standard deviations above the mean of

Test batsmen.

- Pishro-Nik, H. (2014). *Introduction to Probability, Statistics, and Random Processes*. Kappa Research. Section 7.1.2 (central limit theorem; needs a finite variance).

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Z-table (G-2142)	A table that gives, for many values of z, the probability that a standard normal value falls below z, written $\Phi(z)$; used to find normal probabilities.
Standard normal distribution (G-1873)	The normal distribution with mean 0 and standard deviation 1, written $Z \sim N(0,1)$; any normal variable can be converted to it with z-scores, so one table of probabilities serves every normal distribution.
$\phi(z)$ (G-19)	The height of the standard normal bell curve (mean 0, standard deviation 1) at z, which shows how densely values lie near z (its probability density function).
$\Phi(z)$ (G-18)	The probability that a standard normal value is below z: the area under the bell curve to the left of z (the CDF of the standard normal distribution). For example $\Phi(1) = 0.84$.
68-95-99.7 rule (empirical rule) (G-53)	In a column shaped like a bell curve (a normal distribution), about 68.3%, 95.4% and 99.7% of values lie within 1, 2 and 3 standard deviations of the mean.
Z-score (G-2141)	A value after standardization: how many standard deviations it lies from the mean.
Standardization (G-1874)	Scaling a column to mean 0 and standard deviation 1 (subtract the mean, divide by the standard deviation), so features measured on different scales become comparable.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.
Outlier (G-1420)	A value far from the rest of the data.

Term	Meaning
Error (residual) (G-705)	The gap between an actual value and the model's prediction on one data point: actual minus predicted.
Parametric density estimation (G-1451)	Estimating how the data is spread (its probability density) by assuming it follows a named distribution and estimating that distribution's parameters from the data.
Hypothesis testing (G-913)	Checking a claim about a population parameter with a sample.
Central limit theorem (G-364)	For large enough samples, the means of many samples follow a normal distribution centred on the population mean, whatever the shape of the data (if its variance is finite); this justifies confidence intervals and hypothesis tests on means.